1. Launch your IDE(MongoDB Atlas)
2. signup using Organization credentials
3. Create free tier cluster
4. If it is already created, RESUME it.
5. Click on Browse collections
6. Atlas Dashboard → Database → Browse Collections → Add My Own Data.
7. You’re now inside MongoDB Atlas Data Explorer, and you have the sample_mflix database open with the comments collection selected. 8. To run CRUD operations,

a) Create (Insert)

DID the basic things through cmd promt

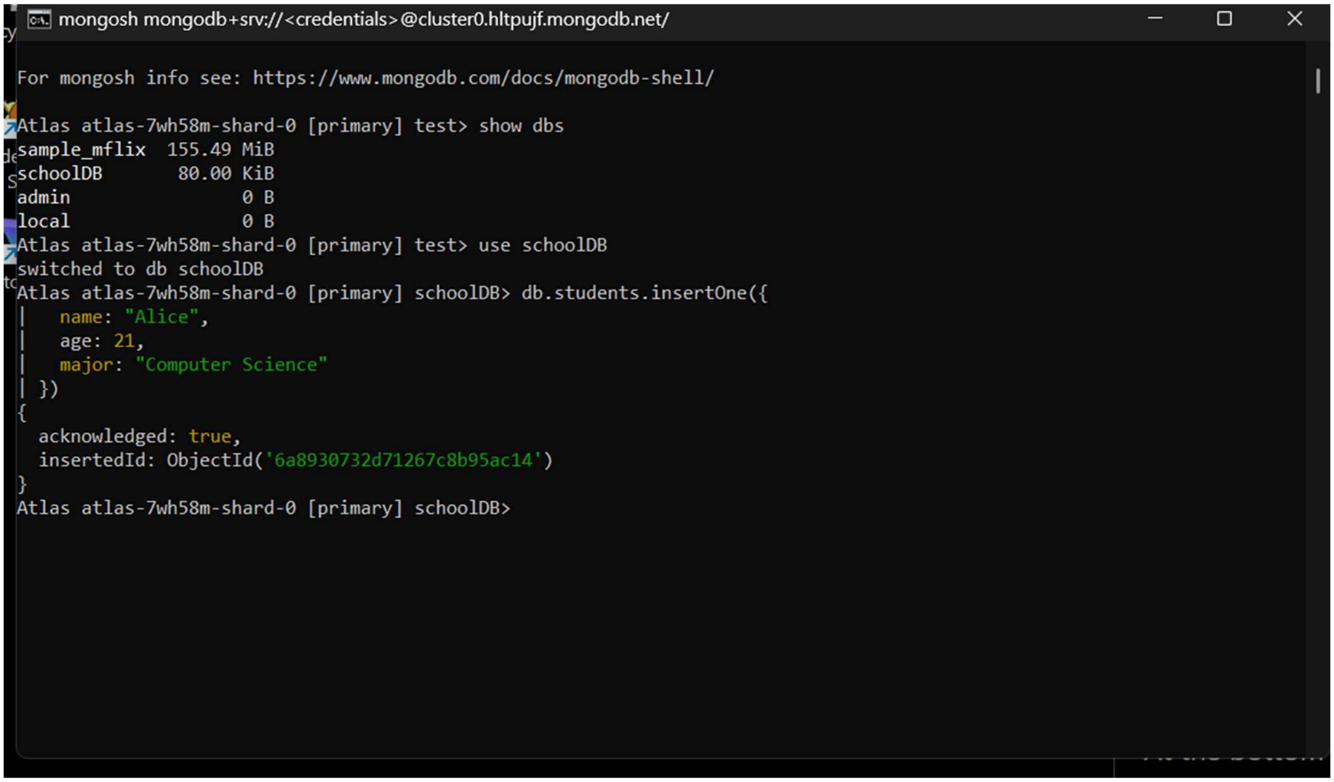
Click “INSERT DOCUMENT” (top-right).

A JSON editor will open. Enter the

details in JSON

```
{
  "name": "Alice",
  "email": "alice@example.com",
  "movie_id": "12345",
  "text": "This is my first review!",
  "date": "2025-08-23T10:00:00Z"
}
```

Click Insert to save.

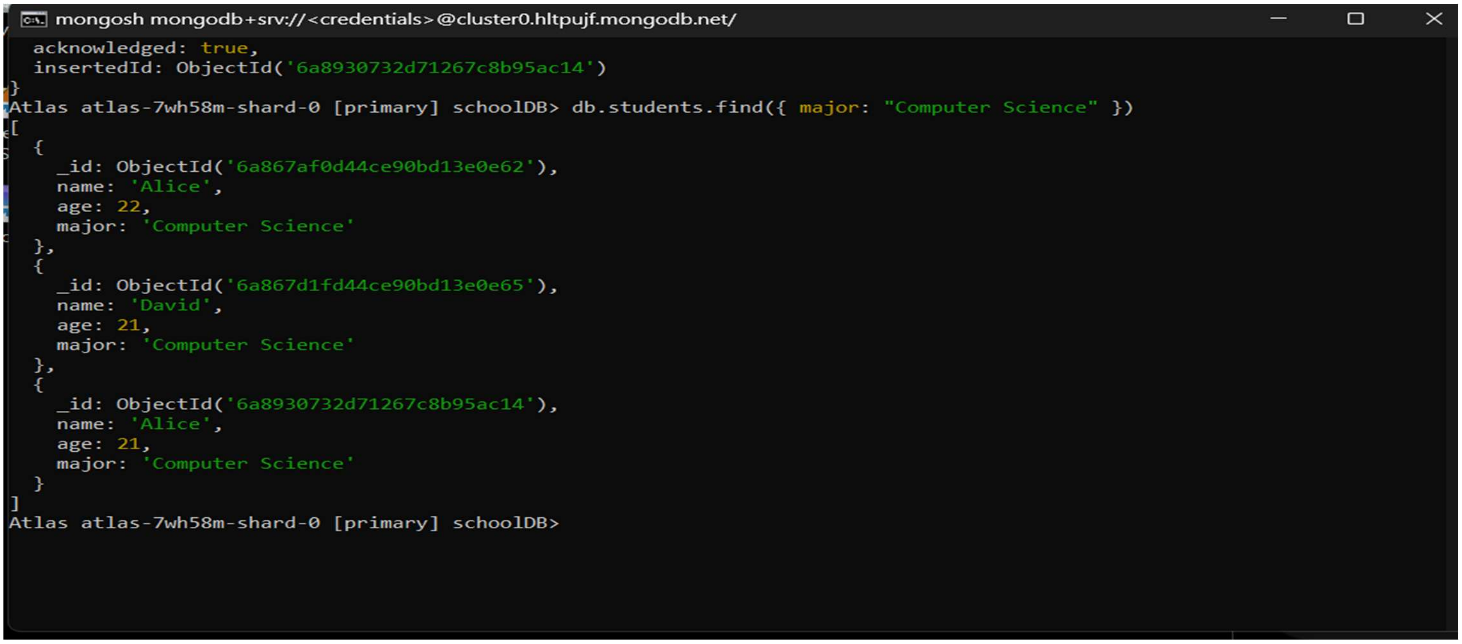


b) Read (Find)

In the Filter box, type queries in JSON format.

Example: Find comments by Alice:

{ "name": "Alice" } click Apply.



c)Update

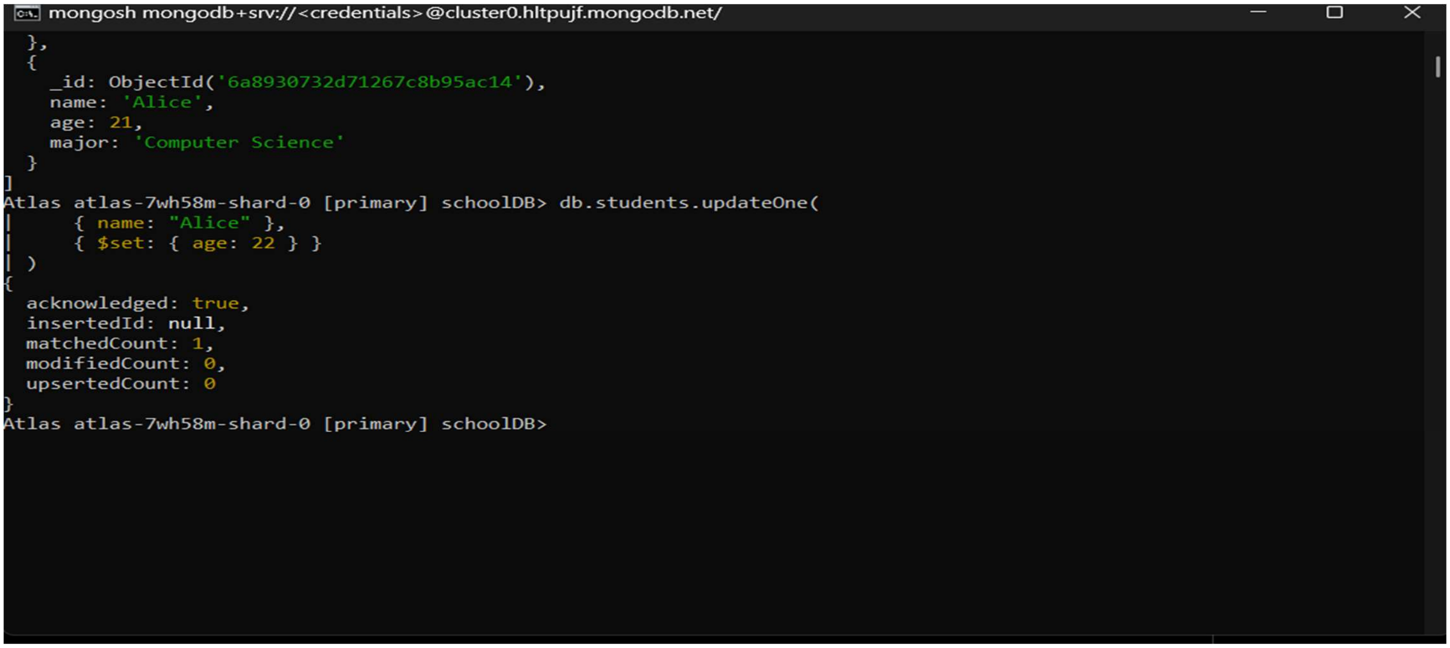
Find the document first using a filter (e.g., { "name": "Alice" }).

Hover over the document → click pencil icon (Edit).

Modify fields, e.g. change "text" to "Updated review!".

Save it.

```
db.students.updateOne(
    { name: "Alice" },
    { $set: { age: 22 } }
)
```

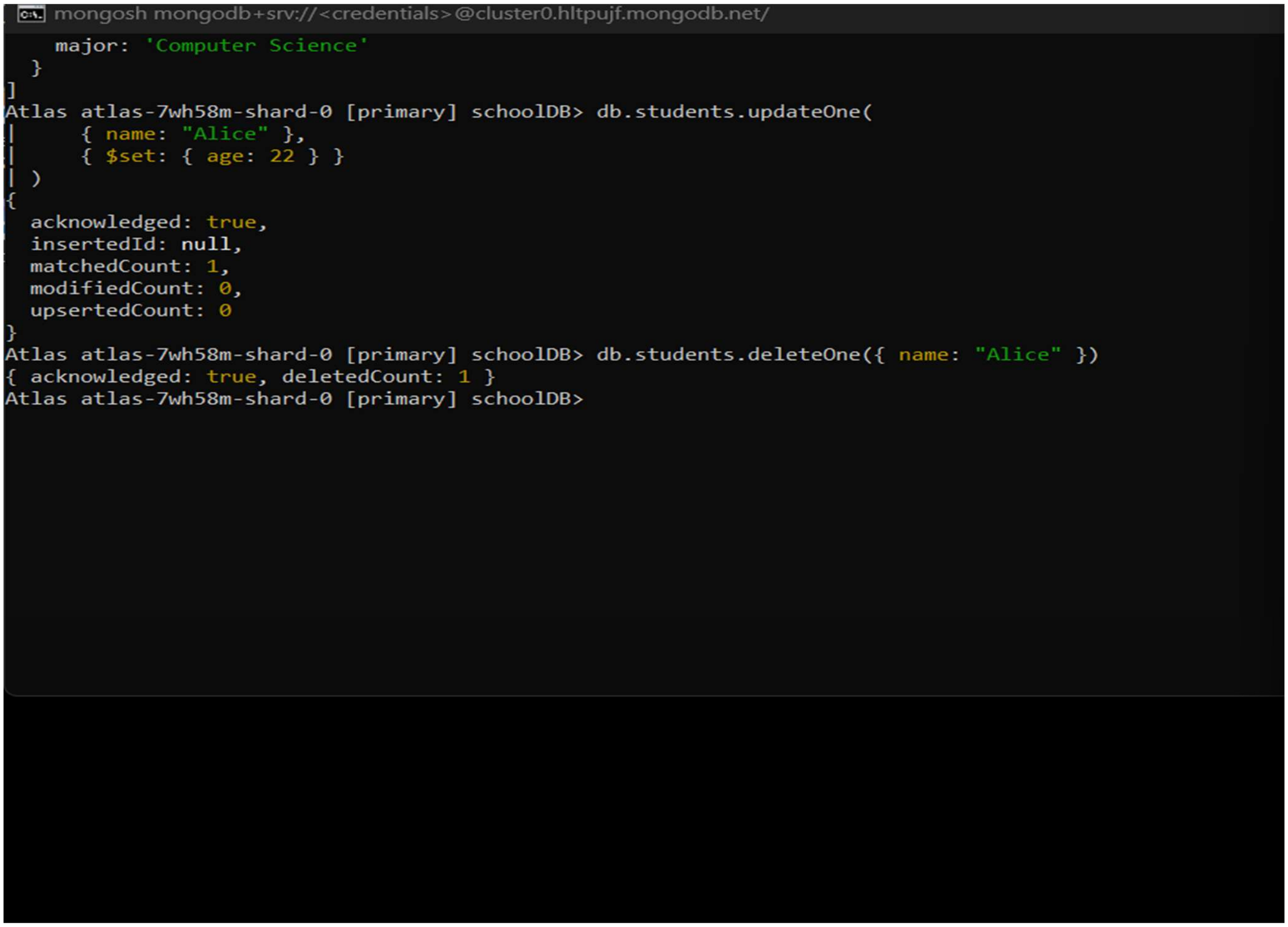


D)Delete

Hover over the document you want to remove.

Click the trash bin icon (Delete).

db.students.deleteOne({ name: "Alice" })



2. Run Queries:

-> use schoolDB
Insert one document,
->db.students.insertOne({ name: "Alice", age: 21, major: "Computer Science" })
Find document,
->db.students.find({ major: "Computer Science" })
Update document,
->db.students.updateOne({ name: "Alice" }, { $set: { age: 22 } })
Delete document,
->db.students.deleteOne({ name: "Alice" })

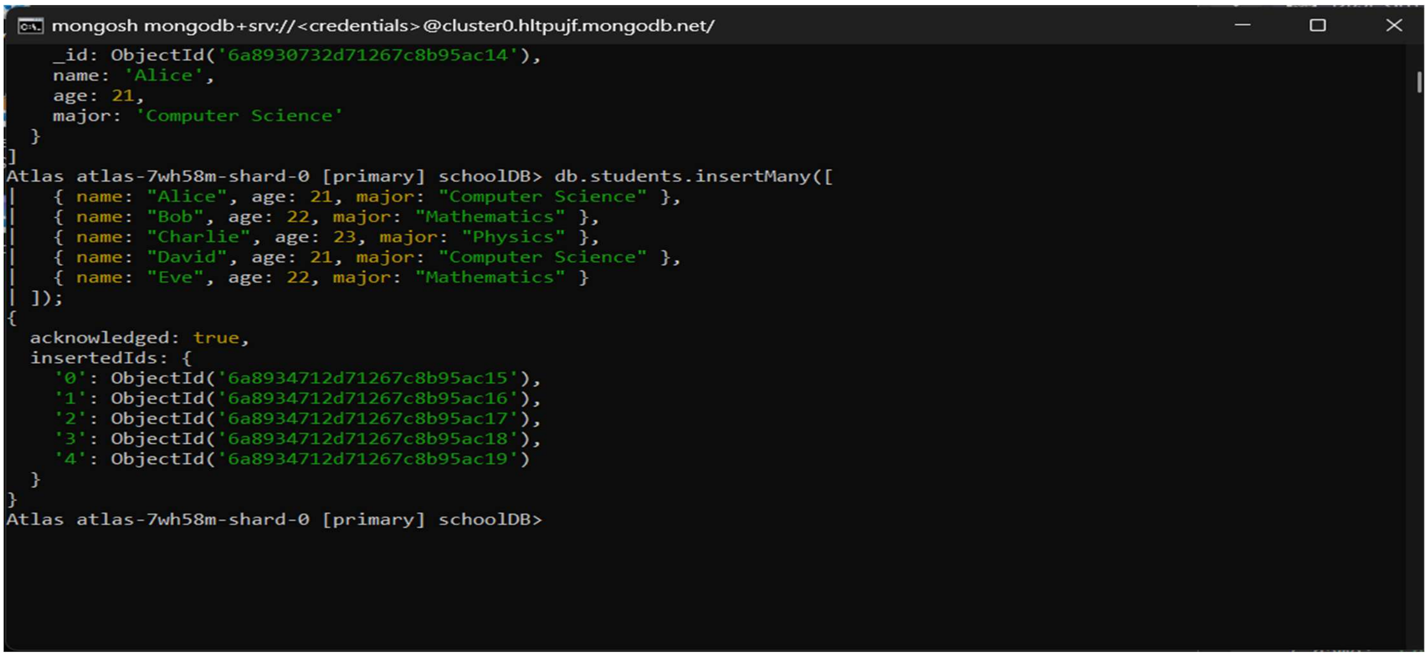
Indexing in MongoDB:

Indexes make searches much faster (just like an index in a book).

Insert many documents in the same database.
db.students.insertMany([
  { name: "Alice", age: 21, major: "Computer Science" },
  { name: "Bob", age: 22, major: "Mathematics" },

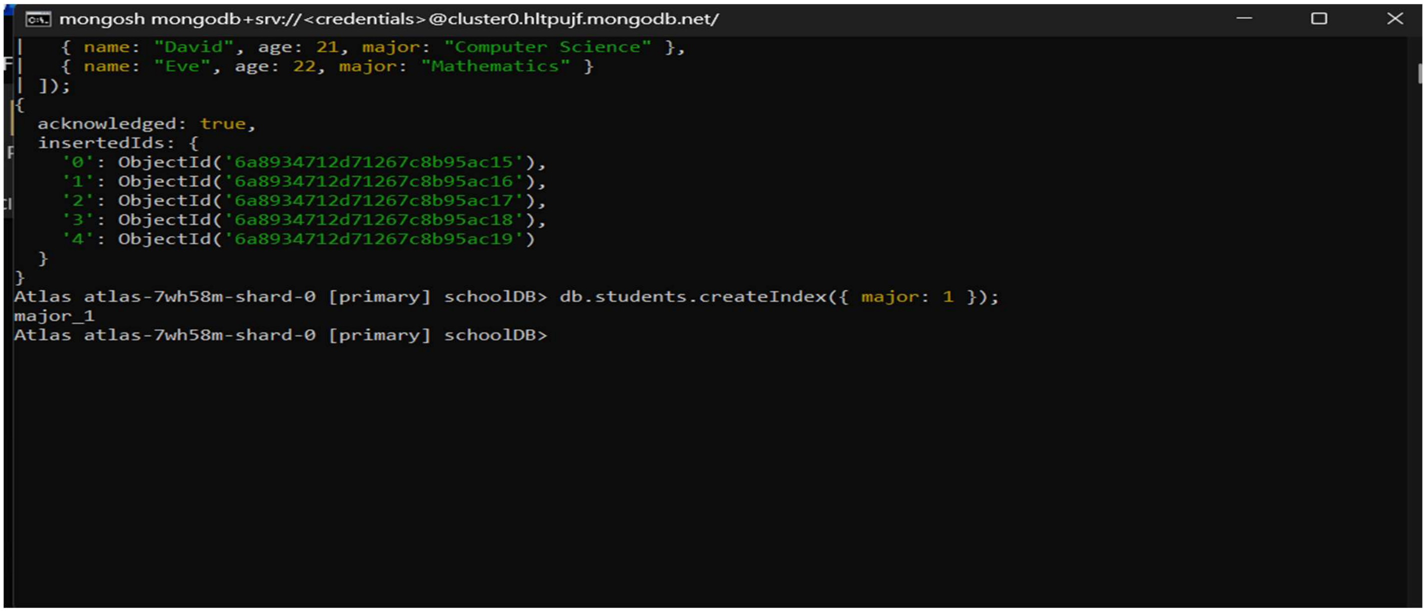
{ name: "Charlie", age: 23, major: "Physics" },
{ name: "David", age: 21, major: "Computer Science" },
{ name: "Eve", age: 22, major: "Mathematics" }
]);

2. Create an Index



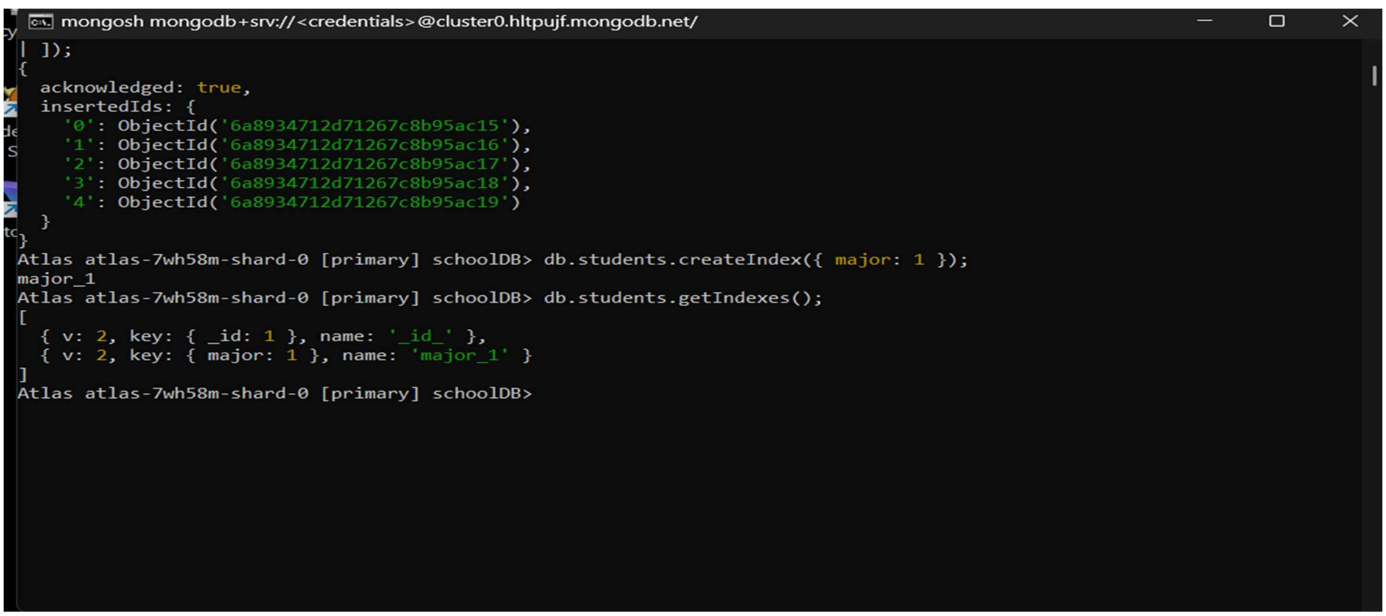
->db.students.createIndex({ major: 1 });

(1 = ascending, -1 = descending)



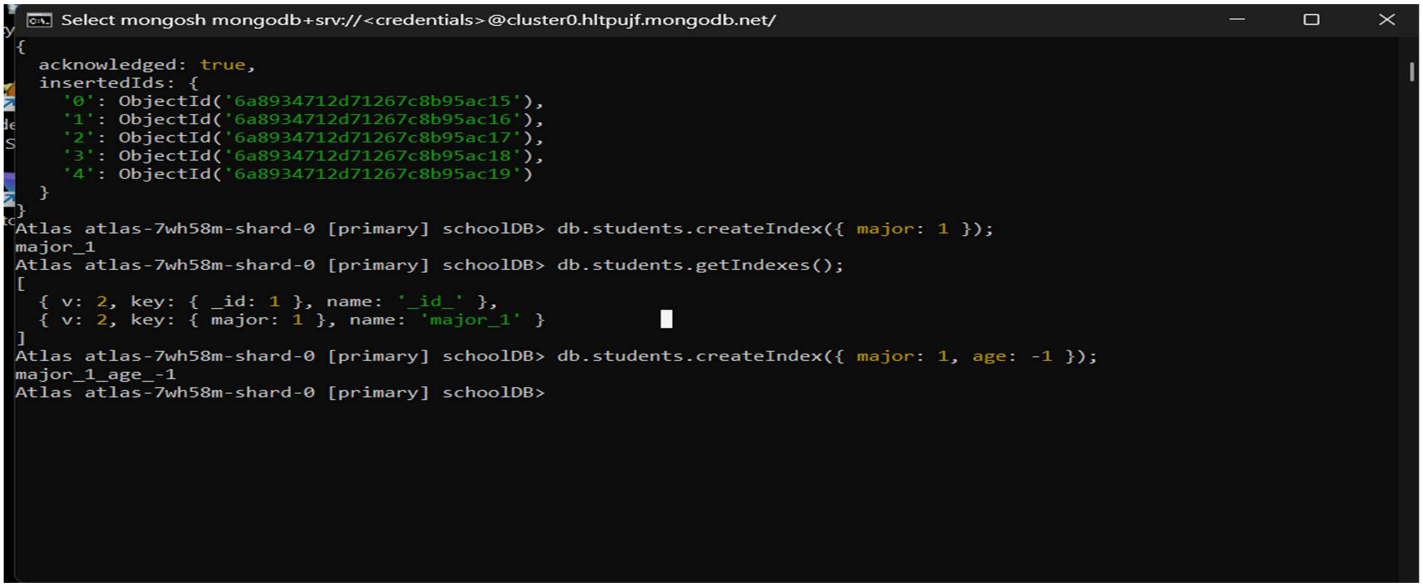
3. View Indexes

-> db.students.getIndexes();



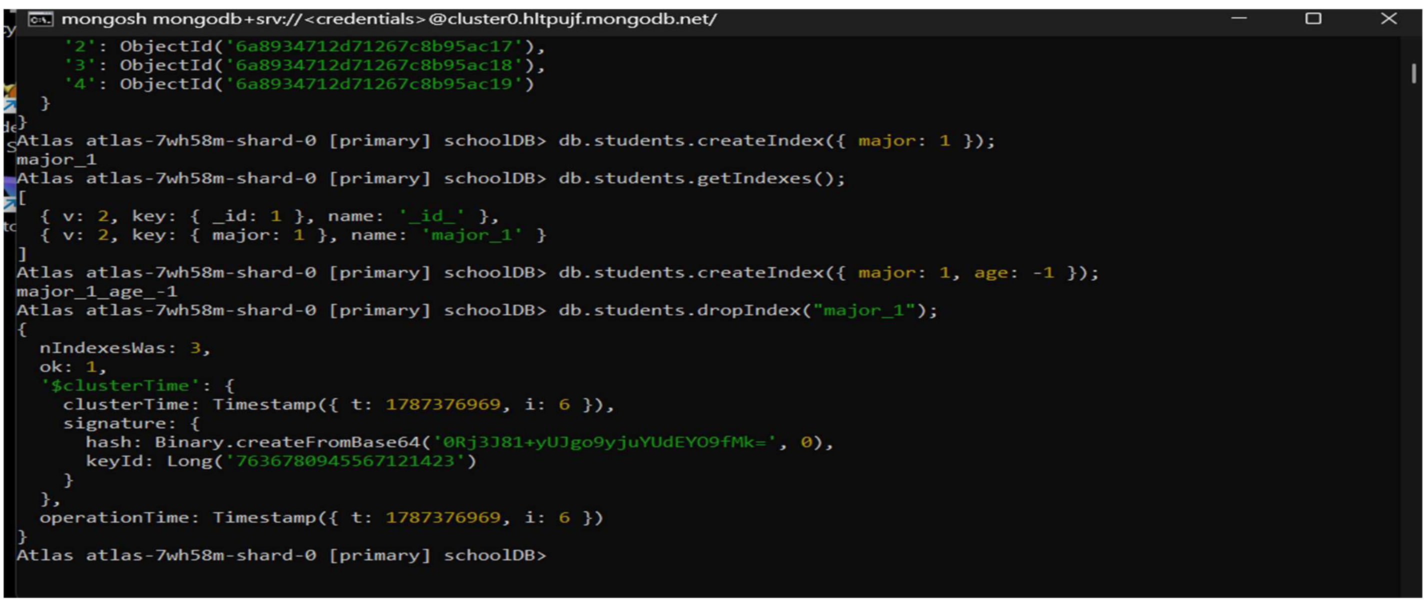
4. Compound Index

-> db.students.createIndex({ major: 1, age: -1 });



5. Drop Index

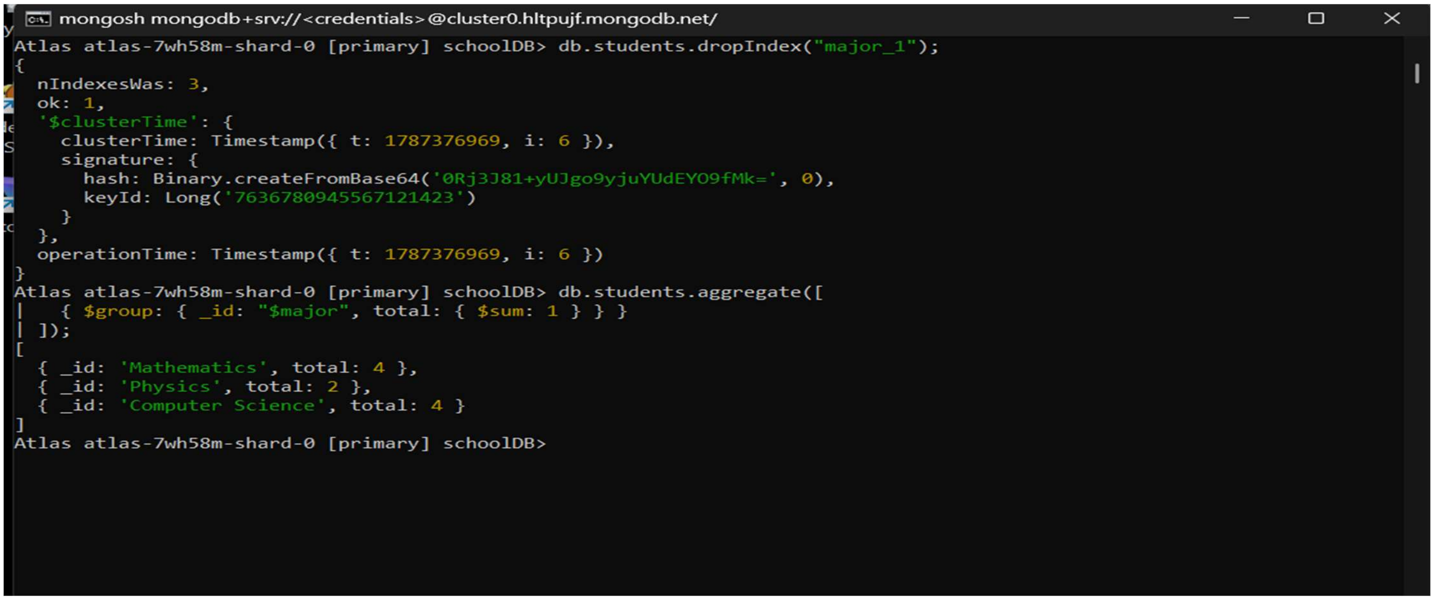
-> db.students.dropIndex("major_1"); Aggregation Framework



in MongoDB:

Count Students by Major

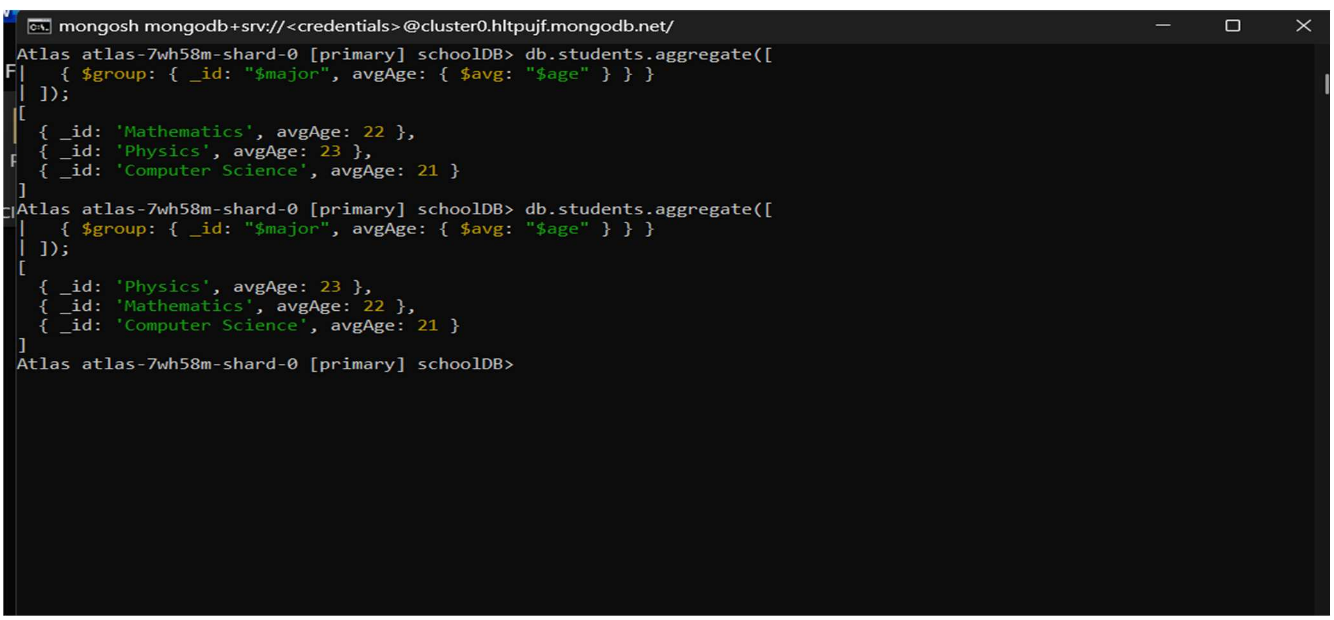
```
-> db.students.aggregate([
  { $group: { _id: "$major", total: { $sum: 1 } } }
]);
```



Groups by major and counts students.

2. Average Age by Major

-> db.students.aggregate([
{ $group: { _id: "$major", avgAge: { $avg: "$age" } } }]);



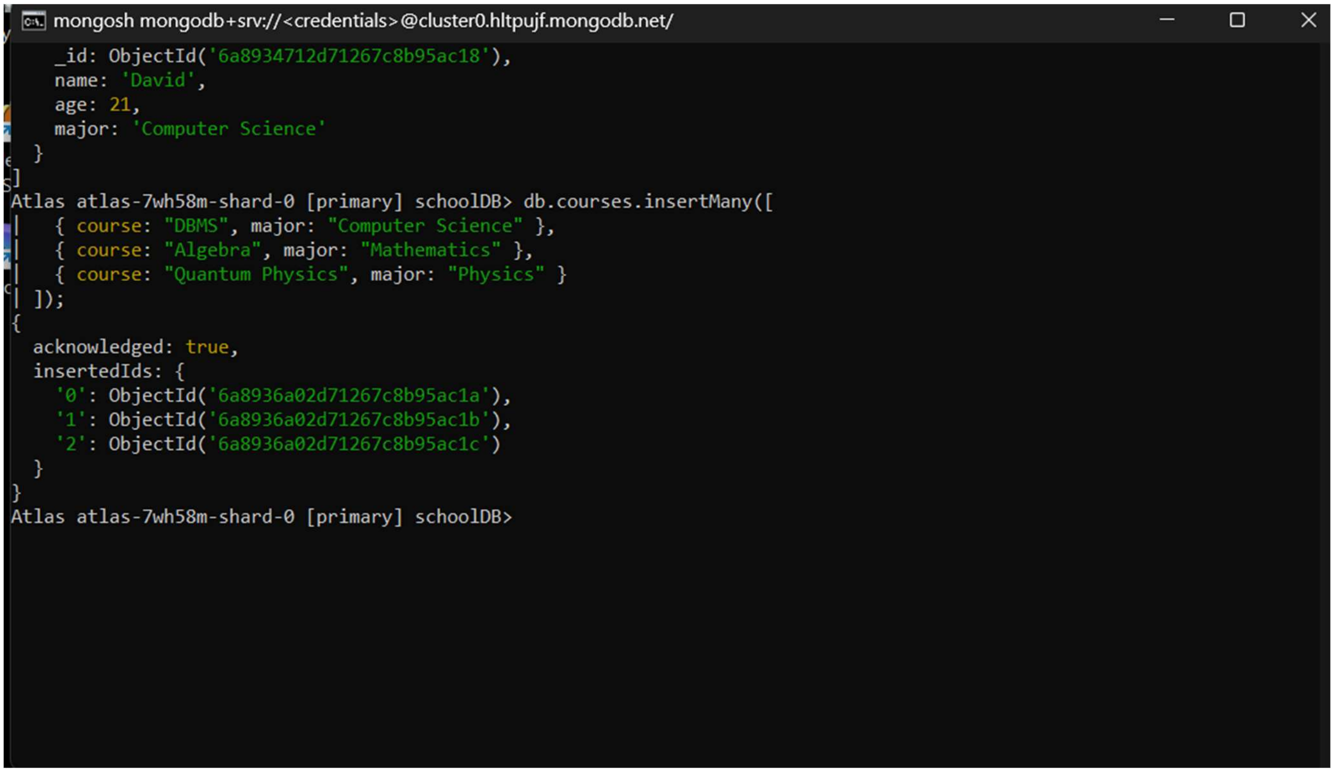
3. sort by Age

-> db.students.aggregate([
{ $sort: { age: -1 } }
]);

```
mongosh mongodb+srv://<credentials>@cluster0.hltpujf.mongodb.net/
  { _id: 'Mathematics', avgAge: 22 },
  { _id: 'Computer Science', avgAge: 21 }
]
Atlas atlas-7wh58m-shard-0 [primary] schoolDB> db.students.aggregate([
|   { $sort: { age: -1 } }
| ]);
[
  {
    _id: ObjectId('6a867d1fd44ce90bd13e0e64'),
    name: 'Charlie',
    age: 23,
    major: 'Physics'
  },
  {
    _id: ObjectId('6a8934712d71267c8b95ac17'),
    name: 'Charlie',
    age: 23,
    major: 'Physics'
  },
  {
    _id: ObjectId('6a867d1fd44ce90bd13e0e63'),
    name: 'Bob',
    age: 22,
    major: 'Mathematics'
  },
  {
    _id: ObjectId('6a867d1fd44ce90bd13e0e66'),
    name: 'Eve',
    age: 22,
    major: 'Mathematics'
```

4. Lookup (Join with another collection) create another collection

```
-> db.courses.insertMany([
  { course: "DBMS", major: "Computer Science" },
  { course: "Algebra", major: "Mathematics" },
  { course: "Quantum Physics", major: "Physics" }
]);
```



Join with students:

```
-> db.students.aggregate([   {
$lookup: {       from: "courses",
localField: "major",       foreignField:
"major",       as: "enrolledCourses"
  }}
]);
```

```
mongosh mongodb+srv://<credentials>@cluster0.hltpujf.mongodb.net/
Atlas atlas-7wh58m-shard-0 [primary] schoolDB> db.students.aggregate([
|   {
|     $lookup: {
|       from: "courses",
|       localField: "major",
|       foreignField: "major",
|       as: "courseInfo"
|     }
|   }
| ]);
[
  {
    _id: ObjectId('6a867d1fd44ce90bd13e0e63'),
    name: 'Bob',
    age: 22,
    major: 'Mathematics',
    courseInfo: [
      {
        _id: ObjectId('6a8936a02d71267c8b95ac1b'),
        course: 'Algebra',
        major: 'Mathematics'
      }
    ]
  },
  {
    _id: ObjectId('6a867d1fd44ce90bd13e0e64'),
    name: 'Charlie',
    age: 23,
    major: 'Physics',
    courseInfo: [
```